

Experiment No. 2

Student Name: Rahul Narwade

PRN: 24070521046

Experiment Title:

Use var, let, const, and demonstrate basic features like template literals and destructuring; build a billing calculator with user input.

Software / Tools Required

- Visual Studio Code
- Google Chrome / Microsoft Edge
- HTML5
- CSS3
- JavaScript (ES6)

Experiment Program Code

Files Used:

- billing.html
- style2.css
- script3.js
- **billing.html**

```
<!DOCTYPE html>
<html>
<head>
<title>Billing Calculator</title>
<link rel="stylesheet" href="style2.css">
</head>

<body>

<div style="
position:fixed;
bottom:15px;
right:15px;
background:#0056b3;
color:white;
padding:10px 15px;
border-radius:10px;
font-size:14px;
font-weight:bold;
box-shadow:0 2px 10px rgba(0,0,0,.3);
z-index:9999;">
Created By:<br>
Rahul Narwade <br>
PRN: 24070521046
</div>

<div class="watermark">
Rahul Narwade
<br>
PRN : 24070521046
</div>

<div class="container">

<h1>■ Billing Calculator</h1>

<input type="text" id="customer" placeholder="Customer Name">

<input type="text" id="item" placeholder="Item Name">

<input type="number" id="price" placeholder="Price">

<input type="number" id="qty" placeholder="Quantity">

<select id="discount">
<option value="0">Discount 0%</option>
<option value="5">Discount 5%</option>
<option value="10">Discount 10%</option>
<option value="15">Discount 15%</option>
<option value="20">Discount 20%</option>
</select>

<select id="gst">
<option value="0">GST 0%</option>
<option value="5">GST 5%</option>
```

```

<option value="12">GST 12%</option>
<option value="18">GST 18%</option>
<option value="28">GST 28%</option>
</select>

<button onclick="generateBill()">Generate Bill</button>

<div class="bill">

<h2>Bill Summary</h2>

<p><b>Customer :</b> <span id="cname">-</span></p>

<p><b>Item :</b> <span id="iname">-</span></p>

<p><b>Price :</b> ■<span id="iprice">0</span></p>

<p><b>Quantity :</b> <span id="iqty">0</span></p>

<p><b>Subtotal :</b> ■<span id="subtotal">0</span></p>

<p><b>Discount :</b> ■<span id="discountAmount">0</span></p>

<p><b>GST :</b> ■<span id="gstAmount">0</span></p>

<hr>

<h2>Total : ■<span id="total">0</span></h2>

</div>

</div>

<script src="script3.js"></script>

</body>
</html>

```

• style2.css

```

*{
margin:0;
padding:0;
box-sizing:border-box;
font-family:Arial,sans-serif;
}

body{
background:linear-gradient(135deg,#36d1dc,#5b86e5);
display:flex;
justify-content:center;
align-items:center;
min-height:100vh;
}

.watermark{
position:fixed;
top:50%;
left:50%;
transform:translate(-50%,-50%) rotate(-30deg);
font-size:42px;
font-weight:bold;
color:rgba(0,0,0,.08);
pointer-events:none;
user-select:none;
text-align:center;
z-index:0;
}

.container{
width:450px;
background:white;
padding:25px;
border-radius:15px;
box-shadow:0 10px 25px rgba(0,0,0,.3);
position:relative;
z-index:1;
}

h1{
text-align:center;
margin-bottom:20px;
color:#0056b3;
}

input,select{
width:100%;
padding:12px;
margin:10px 0;
border-radius:8px;
border:1px solid #ccc;
font-size:16px;
}

button{
width:100%;
padding:12px;

```

```

margin-top:15px;
background:#0056b3;
color:white;
font-size:17px;
border:none;
border-radius:8px;
cursor:pointer;
}

button:hover{
background:#003d80;
}

.bill{
margin-top:20px;
padding:15px;
background:#f4f4f4;
border-radius:10px;
}

.bill p{
margin:8px 0;
}

.bill h2{
text-align:center;
color:#0056b3;
}

hr{
margin:15px 0;
}

```

• script3.js

```

function generateBill(){

let customer=document.getElementById("customer").value;
let item=document.getElementById("item").value;
let price=Number(document.getElementById("price").value);
let qty=Number(document.getElementById("qty").value);
let discount=Number(document.getElementById("discount").value);
let gst=Number(document.getElementById("gst").value);

if(customer=="" || item=="" || price<=0 || qty<=0){
alert("Please fill all details.");
return;
}

let subtotal=price*qty;
let discountAmount=(subtotal*discount)/100;
let amountAfterDiscount=subtotal-discountAmount;
let gstAmount=(amountAfterDiscount*gst)/100;
let total=amountAfterDiscount+gstAmount;

document.getElementById("cname").innerHTML=customer;
document.getElementById("iname").innerHTML=item;
document.getElementById("iprice").innerHTML=price.toFixed(2);
document.getElementById("iqty").innerHTML=qty;
document.getElementById("subtotal").innerHTML=subtotal.toFixed(2);
document.getElementById("discountAmount").innerHTML=discountAmount.toFixed(2);
document.getElementById("gstAmount").innerHTML=gstAmount.toFixed(2);
document.getElementById("total").innerHTML=total.toFixed(2);

}

```

Application Description

The Billing Calculator accepts customer and product details such as customer name, item name, price, quantity, discount, and GST. JavaScript processes the entered values to calculate the subtotal, discount amount, GST amount, and final payable total. The calculated information is then displayed dynamically in the Bill Summary section.

Features Implemented

Feature	Description
User Input	Accepts customer, item, price and quantity details
var, let and const	Demonstrates JavaScript variable declarations
Template Literals	Used to construct formatted bill output
Destructuring	Demonstrates extraction of values from data structures
Billing Calculation	Calculates subtotal, discount, GST and final total
DOM Manipulation	Displays the generated bill dynamically

Feature	Description
Event Handling	Generate Bill button executes JavaScript

Output

Billing Calculator

Customer Name:

Item:

Price:

Quantity:

Discount:

GST:

Generate Bill

Bill Summary

Customer : Rahul Narwade
Item : Tshirt
Price : ₹2234.00
Quantity : 2
Subtotal : ₹4468.00
Discount : ₹223.40
GST : ₹764.03

Total : ₹5008.63

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Output Summary

- Accepts customer and item details through input fields.
- Calculates the subtotal using price and quantity.
- Applies the selected discount percentage.
- Calculates GST on the discounted amount.
- Displays the complete Bill Summary with the final total.
- Demonstrates JavaScript ES6 concepts while processing user input.

Result / Conclusion

The Billing Calculator was implemented successfully using HTML, CSS, and JavaScript. The experiment demonstrated the use of var, let, and const along with ES6 features such as template literals and destructuring. It also demonstrated user input handling, arithmetic calculations, DOM manipulation, and event handling to generate a bill dynamically.